

USB-C 5V Power Board

Design Calculations & Engineering Validation

KiCad 10 project: USB_C_5V_Power_Board

1. USB-C CC Resistors

R1/R2 = 5.1 kΩ Rd pull-downs. Approximate current: $I = 5/5100 = 0.98 \text{ mA}$. Power per resistor: $P = 5^2/5100 = 4.9 \text{ mW}$.

2. Resettable Fuse F1

Littelfuse 1206L050: hold current $\approx 0.50 \text{ A}$ at 20°C ; trip current $\approx 1.00 \text{ A}$ at 20°C . The 0.50 A value is the hold current, not a hard cutoff. Using 0.70Ω maximum resistance, loss at 0.5 A is $I^2R = 0.175 \text{ W}$.

3. TVS Protection D1

Littelfuse SMF5.0A: VRWM = 5.0 V, VBR = 6.4–7.0 V, specified clamp voltage VC = 9.2 V, peak pulse current $\approx 21.7 \text{ A}$. The 9.2 V clamp value applies to the specified surge condition, not normal operation.

4. Reverse-Polarity MOSFET Q1

AO3401A P-channel MOSFET. With gate pulled low, VGS is approximately -5 V . Using RDS(on) max $\approx 60 \text{ m}\Omega$ at VGS = -4.5 V : at 0.5 A , Vdrop $\approx 30 \text{ mV}$ and $P \approx 15 \text{ mW}$.

5. Gate Pull-Down R3

R3 = 100 kΩ. At 5 V: $I = 5/100000 = 50 \mu\text{A}$ and $P = 0.25 \text{ mW}$.

6. 5 V Power Rail

At 0.5 A , idealized 5 V output power is $5 \times 0.5 = 2.5 \text{ W}$. The 5 V and 3.3 V loads share the same USB-C input and protection path.

7. AMS1117-3.3

For 5 V to 3.3 V, voltage drop = 1.7 V. At 0.3 A : Pout = 0.99 W, regulator heat = 0.51 W, idealized efficiency = 66%. At 0.5 A , regulator heat = 0.85 W.

8. AMS1117 Thermal Estimate

Using an illustrative 45°C/W PCB thermal resistance: at 0.3 A , $\Delta T \approx 23^\circ\text{C}$; at 40°C ambient, estimated junction temperature $\approx 63^\circ\text{C}$. At 0.5 A , $\Delta T \approx 38^\circ\text{C}$ and estimated junction temperature $\approx 78^\circ\text{C}$. Actual PCB performance must be measured.

9. Capacitors

C1/C3/C4 = 10 μF , 16 V; C2 = 100 nF, 50 V. Energy $E = \frac{1}{2}CV^2$. A 10 μF capacitor stores 125 μJ at 5 V and about 54.5 μJ at 3.3 V.

10. LED Resistors

Assuming Vf $\approx 2 \text{ V}$: D2/R4 at 5 V with 1 kΩ gives $\approx 3 \text{ mA}$ and $\approx 9 \text{ mW}$ resistor dissipation. D3/R5 at 3.3 V gives $\approx 1.3 \text{ mA}$ and $\approx 1.7 \text{ mW}$. Actual values depend on LED Vf.

11. PCB Trace Considerations

Main power traces are approximately 0.5 mm wide. This is a practical width for a 500 mA-class low-voltage board on typical 1 oz copper, but formal IPC current claims require actual copper thickness, length and temperature-rise limits.

12. Recommended Project Specifications

Parameter	Design value
Input	USB-C, nominal 5 V
USB-C mode	5 V sink only; no USB-PD
CC resistors	5.1 k Ω $\times$ 2
Input protection	1206L050 PTC + SMF5.0A TVS
Reverse polarity	AO3401A P-channel MOSFET
5 V output	Protected 5 V rail
3.3 V regulator	AMS1117-3.3
Recommended 3.3 V continuous target	$\approx$ 300 mA, thermal-dependent
PCB	$\approx$ 50 mm $\times$ 35 mm
Grounding	F.Cu GND copper zone

13. Engineering Limitations

This board is a nominal 5 V USB-C sink and is not USB-PD. Do not advertise 9/12/15/20 V PD input. The 500 mA PTC rating is its hold-current rating at the stated temperature, not a precise cutoff. AMS1117 thermal performance is the main limitation.

14. Validation Status

Schematic ERC: 0 violations. PCB DRC: 0 violations, with intentional USB-C mounting-hole clearance exclusions. Gerbers were generated and inspected. Physical electrical and thermal testing remains required after assembly.